BUSINESS REQUIREMENTS DOCUMENT

Airbnb NYC Listings Analytics

Power BI End-to-End YouTube Demo Project

Data With Pankaj | Version 1.0 | May 2026

Document Title	Business Requirements Document — Airbnb NYC Analytics
Project Name	Airbnb End to End Power BI Demo
Version	1.0 (Initial Release)
Status	Approved for Development
Prepared By	Data With Pankaj (YouTube Channel)
Date	May 2026
Target Audience	Power BI Developers, Data Analysts, YouTube Viewers
Tools	Power BI Desktop, Power Query, DAX, CSV Dataset

1. Executive Summary

This Business Requirements Document (BRD) defines the scope, objectives, data sources, transformation rules, and reporting requirements for the Airbnb NYC Listings Analytics Power BI project. This project is designed as a complete end-to-end YouTube tutorial demonstrating real-world data engineering and business intelligence skills.

The dataset simulates a production-grade Airbnb data extract for New York City, covering 100,000+ booking records from January 2023 through December 2024 across 80 property listings, 50 hosts, and 20 NYC neighbourhoods. The report is designed to answer key business questions around revenue performance, occupancy patterns, host quality, and neighbourhood trends.

> **YouTube Series**
> This project is Part of the 'Data With Pankaj' Power BI end-to-end series. The dataset contains intentional, clearly-documented data quality issues that are fixed step-by-step in Power Query as part of the demonstration.

2. Business Objectives

2.1 Primary Goals

- Enable Airbnb hosts and property managers to monitor revenue performance across NYC neighbourhoods
- Identify high-performing listings and superhosts using occupancy rate, RevPAR, and review score metrics
- Track booking trends over time using month-over-month and year-over-year comparisons
- Demonstrate a complete Power BI development workflow — from raw CSV to published dashboard

2.2 Secondary Goals

- Showcase Power Query M code for real-world data cleaning scenarios
- Demonstrate DAX measure best practices including VAR pattern, CALCULATE, and time intelligence
- Illustrate star schema design principles in Power BI data modeling
- Provide a reusable dataset template for Power BI learners and YouTube viewers

3. Project Scope

3.1 In Scope

- 5 CSV source files: fact_bookings, dim_host, dim_property, dim_location, dim_date
- Power Query data cleaning of 6 documented data quality issues
- Star schema model: 1 fact table, 4 dimension tables
- 24 DAX measures across 5 display folders
- 2-page Power BI report with native visuals only (no custom visuals)
- Dark-theme report with modern UI/UX layout
- YouTube-ready 90-minute demo script

3.2 Out of Scope

- Real-time data refresh or Power BI Gateway configuration
- Row-Level Security (RLS) implementation
- Power BI Service workspace governance and deployment pipelines
- Mobile report optimization beyond basic layout
- Paginated reports (SSRS)

4. Data Sources & Volume

The project uses five CSV files simulating an Airbnb data extract. All files should be placed in a single local folder before loading into Power BI Desktop.

File Name	Table Name	Rows	Columns	Description
fact_bookings.csv	fact_bookings	100,200	13	Central booking transaction table with revenue, reviews, pricing
dim_host.csv	dim_host	50	6	Host profile data — names, since date, response rate, superhost status

File Name	Table Name	Rows	Columns	Description
dim_property.csv	dim_property	80	11	Listing details — room type, property type, amenities, base price
dim_location.csv	dim_location	20	8	NYC neighbourhood lookup with lat/lon and zip code
dim_date.csv	dim_date	1,461	11	Date dimension covering Jan 2023–Dec 2024

5. Data Quality Issues & Power Query Fixes

The following table documents all intentional data quality issues included for the YouTube demonstration. Each issue has a simple, teachable fix using Power Query M functions.

File	Column	Issue	Fix	M Code
dim_host	host_name	3 rows are empty strings (null)	Replace null → 'Unknown Host'	Table.ReplaceValue(...)
dim_host	host_response_rate	3 rows stored as decimal (0.85) instead of '85%'	Detect type, multiply by 100, append %	Table.TransformColumns(...)
dim_property	bedrooms	5 null values for studio-type units	Replace null → 0 (studio)	Table.ReplaceValue(...)
dim_location	latitude	2 values have leading/trailing spaces	Text.Trim → Number.From	Table.TransformColumns(...)
dim_date	month_name	Mixed casing: 'january' vs 'February'	Text.Proper to standardise	Table.TransformColumns(...)
fact_bookings	price	~3,000 rows stored as '$1,234.00' text	Remove $, comma → Decimal	Text.Remove + Number.From
fact_bookings	review_score	~1,500 rows > 5.0 (invalid range)	Filter rows where score ≤ 5.0	Table.SelectRows(...)
fact_bookings	minimum_nights	~800 rows with 0 or negative values	Filter rows where nights ≥ 1	Table.SelectRows(...)
fact_bookings	availability_365	~600 rows with value = 999 (invalid)	Replace 999 → null	Table.ReplaceValue(...)
fact_bookings	booking_id	200 exact duplicate rows	Table.Distinct on booking_id	Table.Distinct(...)

6. Data Model Requirements

6.1 Schema Design

The model follows a classic Star Schema pattern — one central fact table surrounded by four dimension tables. All relationships are one-to-many (1:*) with single-direction filter flow from dimension to fact.

From (Dim)	Key Column	To (Fact)	Cardinality
dim_date	date_key	fact_bookings [date_key]	One-to-Many (1:*)
dim_host	host_id	fact_bookings [host_id]	One-to-Many (1:*)
dim_property	listing_id	fact_bookings [listing_id]	One-to-Many (1:*)
dim_location	location_id	fact_bookings [location_id]	One-to-Many (1:*)

6.2 Modeling Rules

- All FK columns in fact_bookings must be hidden from Report View after relationships are set
- dim_date must be marked as the official Date Table (full_date column)
- All dimension tables must be set to Import mode
- Auto date/time must be disabled (File → Options → Data Load)
- All measures must reside in the '📊 Measures' calculated table

7. Measure Requirements

All 24 measures are organized into 5 display folders within the '📊 Measures' table. Measure names use Title Case with spaces (no underscores).

Display Folder	Measure Name	Format	Business Purpose
💰 Core Revenue KPIs	Total Revenue	$#,0.00	Sum of all booking revenue
💰 Core Revenue KPIs	Total Bookings	#,0	Count of all bookings
💰 Core Revenue KPIs	Avg Nightly Rate	$#,0.00	Average price per night
💰 Core Revenue KPIs	RevPAR	$#,0.00	Revenue per available room
💰 Core Revenue KPIs	Avg Revenue Per Booking	$#,0.00	Revenue ÷ bookings
🏠 Occupancy & Availability	Occupancy Rate	0.0%	(365 - avg avail) / 365
🏠 Occupancy & Availability	Cancellation Rate	0.0%	Strict cancellations / total
🏠 Occupancy & Availability	Avg Nights Per Booking	0.0	Average length of stay
📅 Time Intelligence	Revenue PM	$#,0.00	Prior month revenue
📅 Time Intelligence	MoM Revenue Growth %	+0.0%	Month-over-month growth
📅 Time Intelligence	Revenue PY	$#,0.00	Prior year revenue (SPPLY)
📅 Time Intelligence	YoY Revenue Growth %	+0.0%	Year-over-year growth
📅 Time Intelligence	Revenue YTD	$#,0.00	Year-to-date cumulative

Display Folder	Measure Name	Format	Business Purpose
⭐ Review & Quality	Avg Review Score	0.00	Average guest rating
⭐ Review & Quality	Total Reviews	#,0	Sum of all review counts
⭐ Review & Quality	High Rating %	0.0%	% bookings with score ≥ 4.5
🏆 Host Performance	Superhost Revenue	$#,0.00	Revenue from superhosts only
🏆 Host Performance	Superhost Revenue %	0.0%	Superhost share of total rev
🏆 Host Performance	Superhost Avg Review	0.00	Avg review for superhosts
🏆 Host Performance	Total Active Hosts	#,0	Distinct hosts with bookings
🎨 KPI Formatting	MoM Arrow Label	Text	▲/▼ + % for card display
🎨 KPI Formatting	YoY Arrow Label	Text	▲/▼ + % for card display
🎨 KPI Formatting	Revenue Label	Text	$1.2M compact format

8. Report Requirements

8.1 Page 1 — Executive Overview

- Canvas size: 1280 × 720 px, custom dark background #0F172A
- 4 KPI cards: Total Revenue, Avg Nightly Rate, Occupancy Rate, Avg Review Score
- Line chart: Monthly revenue trend, 3 years overlaid (2023, 2024, 2025)
- Clustered bar: Top 10 neighbourhoods by revenue
- Donut chart: Room type revenue split
- Slicers: Year (horizontal tile), Room Type (vertical list)
- All visuals use native Power BI visuals only — no custom visual imports

8.2 Page 2 — Host & Property Drill

- Bubble/Scatter chart: Avg Nightly Rate (X) vs Avg Review Score (Y), bubble size = Revenue
- Clustered bar: Superhost vs Regular host comparison across Revenue, Occupancy, Reviews
- Matrix/Table: Host performance table with data bar conditional formatting on Revenue column
- Decomposition tree: Revenue breakdown by Neighbourhood → Room Type → Superhost flag
- Page navigation button linking back to Page 1

9. Acceptance Criteria

- All 5 CSV files load without errors in Power Query
- All 10 data quality issues are resolved before Close & Apply
- Star schema model shows 4 active relationships with correct cardinality
- 24 measures exist in the '📊 Measures' table across 5 folders
- Total Revenue measure returns a plausible non-zero value when no filters are applied

- Time intelligence measures (MoM, YoY, YTD) return non-blank values when date slicer is set
- Page 1 visuals cross-filter correctly when a neighbourhood bar is clicked
- Report renders correctly in both light and dark Power BI theme

10. Document Sign-Off

Prepared By	Data With Pankaj
Reviewed By	Pankaj Namekar — Senior Power BI Developer
Approved By	Data With Pankaj Channel
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Approval Date	May 2026
Next Review	On project completion / post-YouTube publish

By proceeding with this project, all stakeholders acknowledge the scope, data sources, and acceptance criteria defined in this document.